

Intelligent Attack Detection System using NMS Data, LLMs, and Interactive Dashboard

Major Project (CS1239)

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1. Introduction:

With rapid growth of networked systems, the need for security and stability of networks has become a critical concern. Network Management Systems (NMS) provides detailed telemetry (process of recording), including metrics such as bandwidth usage, latency, packet drops and data logs. But these raw metrics alone are not sufficient for proactive attack detection. This project aims to design and implement a pipeline that fetches real-time data from an NMS API (in our case Zabbix) and the preprocess and train it is using machine learning/LLMs to detect anomalies or the potential attacks, and at last visualizes results in a user-friendly dashboard for ease of use to network administrators.

2. Objectives:

- a. **Data Collection:** Firstly, we must fetch data and aggregate real-time metrics and logs from the NMS using APIs.
- b. **Preprocessing:** Then we must clean the data, normalize it and then transform the raw network data into structure datasets.
- c. **Model Development:** Fine-tune a LLM (or Train a ML) model on the created dataset to identify whether an attack is being done on the network.
- d. **Decision System:** Then automate detection and generate alerts when anomalies are detected.
- e. **Dashboard Development:** Build an interactive dashboard for real-time monitoring, visualization of metrics, and attack alerts.

3. Methodology:

- a. Data Acquisition
- b. Data Processing
- c. Model Building
- d. Attack Detection System
- e. Dashboard Implementation

4. Expected Outcomes:

Our project aims to develop a fully functional system that can do the following:

- Fetches data from a live network using NMS APIs
- Identifies anomalies and such potential cyberattacks which may harm an organization with high accuracy

- Provides actionable insights via a dashboard

Help network administrators respond quickly to the incidents, reducing downtime and mitigating threats.

5. Tools & Technologies:

- Network Monitoring: Zabbix, FastAPI, Uvicorn, PostgreSQL
- Model Development: TensorFlow, HuggingFace
- Dashboard Development: React, FastAPI

6. Applications:

This project can have various applications, some of which are stated below:

- ISPs to detect anomalies in customer traffic
- Enterprise networks for proactive attack monitoring
- Research in network security anomaly detection
- Data centers and cloud providers for automated intrusion detection

7. Conclusion:

This project brings together network data analytics, LLMs, and interactive visualization to build a robust platform for intelligent attack detection. By using real-time NMS data and applying predictive modeling, the system can accurately identify anomalies and potential cyberattacks. The integration of automated detection with a user-friendly dashboard ensures that network administrators can monitor traffic seamlessly and take quick corrective measures against that traffic. This proactive approach reduces the risk of system downtime and enhances the resilience of network infrastructure.

Therefore, this solution contributes toward building smarter, more efficient, secure and self-sustaining networks in the era of evolving cyber threats.